

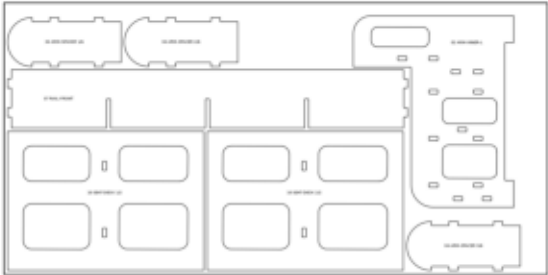
LENA SOFA / CNC NOTES

4 sheets 2440 x 1220 x 18 mm (sheet size ASSUMED)

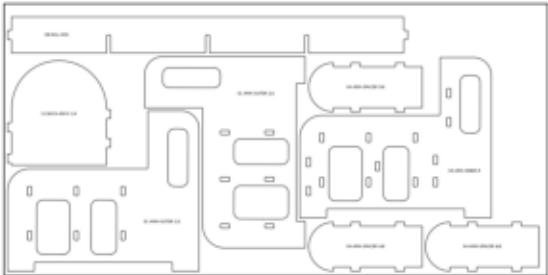
Units / scale	millimetres, 1:1 - never rescale the DXF
Layers	CUT (all closed, cut through) / ENGRAVE-LABEL (0.5-1 mm) / REFERENCE-SHEET (do not cut)
Joints	through-mortises 41 x 19 mm for 40 mm tabs in 18 mm board - UNTESTED FIT
Corner relief	R3.2 dogbone in every inside corner, sized for a 6 mm bit (ASSUMED)
Cutter offset	apply the normal tool-radius offset on every CUT contour; no extra compensation inside slots
Cut order	engrave, then inside contours (mortises, openings), then outlines
Hold-down	parts 10 mm apart, 9 mm clamp edge; add onion-skin or tabs for small ribs
Small parts	spacers and arches sit next to long rails - add tabs or onion-skin so they cannot move on the last pass

LENA SOFA / CUTTING LAYOUT

SHEET 1/4 - 2440 x 1220 x 18 mm MDF



SHEET 2/4 - 2440 x 1220 x 18 mm MDF



SHEET 3/4 - 2440 x 1220 x 18 mm MDF



SHEET 4/4 - 2440 x 1220 x 18 mm MDF



LENA SOFA / VALIDATION

automated checks - geometry, joints, 3D assembly

PASS	[1] part topology	01 RIB-A: 8 loops, simple, holes inside and apart
PASS	[1] part topology	02 RIB-B: 8 loops, simple, holes inside and apart
PASS	[1] part topology	03 BACK-ARCH: 1 loops, simple, holes inside and apart
PASS	[1] part topology	04 RAIL-FRONT: 1 loops, simple, holes inside and apart
PASS	[1] part topology	05 RAIL-MID: 1 loops, simple, holes inside and apart
PASS	[1] part topology	06 RAIL-BACK: 1 loops, simple, holes inside and apart
PASS	[1] part topology	07 ARM-INNER-L: 17 loops, simple, holes inside and apart
PASS	[1] part topology	08 ARM-INNER-R: 17 loops, simple, holes inside and apart
PASS	[1] part topology	09 ARM-SPACER: 1 loops, simple, holes inside and apart
PASS	[1] part topology	10 ARM-OUTER: 10 loops, simple, holes inside and apart
PASS	[1] part topology	11 SEAT-DECK: 7 loops, simple, holes inside and apart
PASS	[2] joints (Tokyo rule: 19 x 41 mortise for 18 mm board, 40 mm tab)	mortise 41 x 19 mm, tab 40 mm, board 18 mm
PASS	[2] joints (Tokyo rule: 19 x 41 mortise for 18 mm board, 40 mm tab)	54 mortises, all 19 x 41 mm: {(19.0, 41.0): 54}
PASS	[2] joints (Tokyo rule: 19 x 41 mortise for 18 mm board, 40 mm tab)	relief R3.2 clears a 6 mm cutter
PASS	[2] joints (Tokyo rule: 19 x 41 mortise for 18 mm board, 40 mm tab)	thinnest web beside a post mortise 19.0 mm (>= 18 mm)
PASS	[2] joints (Tokyo rule: 19 x 41 mortise for 18 mm board, 40 mm tab)	rail cross-halvings keep 81 mm of rail and rib each side
PASS	[3] corner relief and cutter reach	01 RIB-A: 13 corners relieved, worst spot a 6 mm bit cannot reach 0.01 mm2
PASS	[3] corner relief and cutter reach	02 RIB-B: 9 corners relieved, worst spot a 6 mm bit cannot reach 0.01 mm2
PASS	[3] corner relief and cutter reach	03 BACK-ARCH: 8 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	04 RAIL-FRONT: 14 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	05 RAIL-MID: 10 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	06 RAIL-BACK: 14 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	07 ARM-INNER-L: 3 corners relieved, worst spot a 6 mm bit cannot reach 0.01 mm2
PASS	[3] corner relief and cutter reach	08 ARM-INNER-R: 3 corners relieved, worst spot a 6 mm bit cannot reach 0.01 mm2
PASS	[3] corner relief and cutter reach	09 ARM-SPACER: 10 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	10 ARM-OUTER: 3 corners relieved, worst spot a 6 mm bit cannot reach 0.01 mm2
PASS	[3] corner relief and cutter reach	11 SEAT-DECK: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.01 mm2
PASS	[4] nesting sheet 1: 7 parts, closest 12.0 mm apart, all inside the 9 mm edge	
PASS	[4] nesting sheet 2: 8 parts, closest 14.0 mm apart, all inside the 9 mm edge	
PASS	[4] nesting sheet 3: 5 parts, closest 14.9 mm apart, all inside the 9 mm edge	
PASS	[4] nesting sheet 4: 2 parts, closest 15.7 mm apart, all inside the 9 mm edge	
PASS	[4] nesting	every part placed its full quantity (22 pieces)
PASS	[5] labels	22 pieces labelled
PASS	[5] labels	every label sits on material, 4 mm from any cut
PASS	[6] built DXF	units = mm
PASS	[6] built DXF	105 closed CUT profiles (expected 105)
PASS	[6] built DXF	22 ENGRAVE-LABEL texts
PASS	[6] built DXF	no duplicate CUT profiles (0 found)
PASS	[6] built DXF	scale 1:1 (front rail 1796.0 mm in model and in the DXF)
PASS	[6] built DXF	longest part 1796 mm fits the sheet
PASS	[3D-1] interference	42 neighbouring pairs, 0 interpenetrate
PASS	[3D-2] every mortise holds a tab	54/54 mortises filled
PASS	[3D-3] load path	on the floor: ARM-INNER-L, ARM-INNER-R, ARM-OUTER, RIB-A, RIB-B
PASS	[3D-3] load path	6/6 spacers bear on both panels of their arm
PASS	[3D-3] load path	3/3 rails sit in all 3 ribs and both arm panels
PASS	[3D-3] load path	2/2 decks bear on 2 ribs and all 3 rails
PASS	[3D-3] load path	4/4 arches tabbed into the posts either side
PASS	[3D-4] assembly sequence:	every piece slides home along one axis 22 pieces in order, 0 blocked on the way in
PASS	[3D-5] overall size	frame 2200.0 x 860.0 x 800.0 mm, standing on z = 0